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Keywords: Standard Model Effective Field Theory, Future Colliders, Parton Distribution Functions, Deep Learning, Likelihood-Free Inference

Education

- **University of Edinburgh** Edinburgh, United Kingdom
  Postdoctoral Researcher *since Oct 2024*
  - Postdoctoral Researcher in Theoretical Particle Physics at the Higgs Centre for Theoretical Physics
- **VU Amsterdam** Amsterdam, The Netherlands
  PhD candidate *Oct 2020 - Sep 2024*
  - PhD candidate in Theoretical Particle Physics at VU Amsterdam and Nikhef
    Advisors: Juan Rojo, Wouter Verkerke
- **Utrecht University** Utrecht, The Netherlands
  M.Sc. Theoretical Physics *Sep 2018 - Jul 2020*
  - Thesis on 'Matching between EFT and UV Complete Models' at Nikhef
    Graded 8.5/10.0
- **Utrecht University** Utrecht, The Netherlands
  B.Sc. Physics & Astronomy *Sep 2015 - Jun 2018*
  - Graduated cum laude, GPA 4.0/4.0 (Dutch grading system: 8.75/10.0)
  - Thesis on 'Renormalization Group Connected to Neural Networks' at the Institute of Theoretical Physics (Utrecht)
- **University of Edinburgh** Edinburgh, United Kingdom
  Exchange programme *Sep 2017 - Dec 2017*
  - Obtained an average grade of 80 %, 1st Class Honours

Publications

- R. D. Ball, **J. ter Hoeve**, T. Sharma, R. Stegeman, *A simultaneous determination of $\alpha_s(m_Z)$ and m_t from a global PDF analysis*, in preparation
- T. Armadillo et al., *SMEFiT Projections for Modified Couplings, SMEFT Operators and UV-Complete Scenarios at Future Particle Colliders*, in preparation
- J. Altmann et al., *ECFA Higgs, electroweak, and top Factory Study*, vol. 5/2025 of *CERN Yellow Reports: Monographs*. 6, 2025, 10.23731/CYRM-2025-005, [2506.15390].
- FCC collaboration, *Future Circular Collider Feasibility Study Report: Volume 1, Physics, Experiments, Detectors*, 2505.00272.
- FCC collaboration, *Future Circular Collider Feasibility Study Report: Volume 2, Accelerators, Technical Infrastructure and Safety*, 2505.00274.

- FCC collaboration, *Future Circular Collider Feasibility Study Report: Volume 3, Civil Engineering, Implementation and Sustainability*, 2505.00273.
- **J. ter Hoeve**, L. Mantani, J. Rojo, A. N. Rossia and E. Vryonidou, *Higgs trilinear coupling in the standard model effective field theory at the high luminosity LHC and the FCC-ee*, *Phys. Rev. D* **112** (2025) 013008 [2504.05974].
- **J. ter Hoeve**, *Fingerprinting New Physics with Effective Field Theories*, Ph.D. thesis, 2025. 2501.08818.
- **J. ter Hoeve**, L. Mantani, J. Rojo, A. N. Rossia and E. Vryonidou, *Connecting scales: RGE effects in the SMEFT at the LHC and future colliders*, *JHEP* **06** (2025) 125 [2502.20453].
- E. Celada, T. Giani, **J. ter Hoeve**, L. Mantani, J. Rojo, A. N. Rossia et al., *Mapping the SMEFT at high-energy colliders: from LEP and the (HL-)LHC to the FCC-ee*, *JHEP* **09** (2024) 091 [2404.12809].
- **J. ter Hoeve**, E. Laenen, C. Marinissen, L. Vernazza and G. Wang, *Region analysis of QED massive fermion form factor*, *JHEP* **02** (2024) 024 [2311.16215].
- **J. ter Hoeve**, G. Magni, J. Rojo, A. N. Rossia and E. Vryonidou, *The automation of SMEFT-assisted constraints on UV-complete models*, *JHEP* **01** (2024) 179 [2309.04523].
- H. La, A. Brokkelkamp, S. van der Lippe, **J. ter Hoeve**, J. Rojo and S. Conesa-Boj, *Edge-induced excitations in Bi2Te3 from spatially-resolved electron energy-gain spectroscopy*, *Ultramicroscopy* **254** (2023) 113841 [2305.03752].
- R. Gomez Ambrosio, **J. ter Hoeve**, M. Madigan, J. Rojo and V. Sanz, *Unbinned multivariate observables for global SMEFT analyses from machine learning*, *JHEP* **03** (2023) 033 [2211.02058].
- A. Brokkelkamp, **J. ter Hoeve**, I. Postmes, S. E. van Heijst, L. Maduro, A. V. Davydov et al., *Spatially Resolved Band Gap and Dielectric Function in Two-Dimensional Materials from Electron Energy Loss Spectroscopy*, *J. Phys. Chem. A* **126** (2022) 1255 [2202.12572].

Seminars and talks at international conferences

- *Towards a combined α_s and m_t determination from a global PDF analysis*, EPS-HEP, July 2025, (slides)
- *Connecting scales: RGE effects in the SMEFT at the LHC and future colliders*, EPS-HEP, July 2025, (slides)
- Contributed to E. Laenen's opening talk at the Venice Open Symposium, June 2025, (slides, nr. 19)
- *Connecting scales in the SMEFT at the LHC and future colliders*, Seminar at DAMTP, University of Cambridge, May 2025, (slides)
- *Single particle extensions at FCC-ee*, 8th FCC Physics Workshop, CERN, Jan 2025, (slides)
- *Mapping the SMEFT at future colliders, towards the ESPPU*, Seminar at IPPP, Durham University, Dec 2024
- *Mapping the SMEFT at High-Energy Colliders: from LEP and the (HL-)LHC to the FCC-ee*, ICHEP, Prague, Jul 2024, (slides)

- *Update on global SMEFT fits*, invited talk at Standard Model at the LHC, Rome, May 2024, (slides)
- *Global SMEFT fits from (HL)-LHC to future colliders*, Seminar Radboud University, Nijmegen, March 2024
- *Likelihood-free inference for global SMEFT fits*, invited talk at the 1st COMETA General Meeting, Izmir, Feb 2024, (slides)
- *Machine Learning opportunities for EFT analyses*, invited talk at the LHC EFT Working Group, CERN (online), Oct 2023, (slides), (recording)
- *Likelihood learning theory in practice*, invited talk at the LHC Precision Program, Benasque (Spain), Oct 2023, (slides)
- *Interpreting HEP data in SMEFiT*, invited talk at the Reinterpretation Forum, Durham, Aug 2023, (slides)
- *Statistically optimal observables for global SMEFT fits*, HEFT, Manchester, Jun 2023, (slides)
- *Unbinned multivariate observables for global SMEFT analyses from machine learning*, invited talk at CERN, Geneva, Dec 2022, (slides)
- *Towards an optimal global SMEFT fit with machine learning*, Machine Learning at GGI, Florence, Sep 2022, (slides)
- *Unbinned measurements in global SMEFT fits from machine learning*, Learning to Discover, Paris-Saclay, May 2022, (slides)
- *Charting Electron Energy Loss Spectra with Machine Learning*, NWO Physics 2021, Jan 2022
- *Optimally sensitive observables for global EFT fits*, Dutch Physics Society (NNV) annual meeting, Nov 2021, (slides)
- *Optimally sensitive observables for global EFT fits*, Higgs 2021, Stony Brook University (online), Oct 2021, (slides)

Software packages

- `SMEFiT`, a Standard Model Effective Field Theory Fitter
  `smefit.science` (open source)
- `NNPDF` Neural Network Parton Distribution Functions
  `docs.nnpdf.science` (open source)
- `ML4EFT`, likelihood-free inference tool based on deep-learning for particle physics data
  `lhcfitnikhef.github.io/ML4EFT` (open source)
- `EELSFitter`, an Electron-Energy Loss Spectra analyser
  `lhcfitnikhef.github.io/EELSfitter` (open source)

Referee papers

- **SciPost**: G. Guedes and P. Olgoso, *From the EFT to the UV: the complete SMEFT one-loop dictionary*, 2412.14253.
- **EPJC**: D. Keller, *Experimental Uncertainty Propagation in Neural Network Extraction in Hadronic Physics*, 2509.11456.
- **SciPost**: S. Degenkolb, N. Elmer, T. Modak, M. Mühlleitner and T. Plehn, *A Global View of the EDM Landscape*, 2403.02052.
- **JHEP**: S. Chai, J. Gu and L. Li, *From optimal observables to machine learning: an effective-field-theory analysis of $e^+e \to W^+W$ at future lepton colliders*, *JHEP* **05** (2024) 292 [2401.02474].

Activities and outreach

- **SMEFT workshop** — Edinburgh
  Co-organised SMEFT workshop at the Higgs Centre — *Nov 2024*
- **NNV subatomic physics council member**
  National Physics Society — *Jun 2022 - Sep 2024*
- **Chair DRSTP PhD council**
  Dutch Research School for Theoretical Physics — *Dec 2020 - Sep 2024*
  - Organise scientific meetings and social/outreach events for theoretical physics PhD students in The Netherlands
- **Outreach talk** — Amsterdam
  Norwegian high school students — *April 2024*
- **PhysTev Les Houches workshop** — Les Houches, France
  International workshop aiming to push the frontier of particle physics — *June 2023*
- **CERN Masterclass** — Nikhef
  Particle Physics outreach to high school students — *March 2022*
- **Invited guest lecture** — TU Delft
  Topic: Electron Energy Loss Fitter with Machine Learning — *May 2021*

Teaching and supervisory experience

- **PhD student supervision** — Edinburgh
  A. Pietraszkiewicz — *since Sep 24*
- **Master thesis supervision** — joint UvA/VU
  P. Herbschleb, MSc Theoretical Physics, optimal observables in SMEFT — *Sep 22- Jun 23*
- **Bachelor thesis supervision** — joint UvA/VU
  W. Gauthier (2023), J. Bakker and D. Pelan (2021), BSc Physics & Astronomy
- **Teaching Assistant** — joint UvA/VU
  Courses: — *since Sep 2020*

  - Effective Field Theories (2023)
  - Quantum Mechanics II (2021, 2022)

- **Teaching Assistant** Utrecht University
  Courses: *Nov 2016 - Jun 2020*
  - Black Holes (2019, 2020)
  - Fluid Dynamics (2018, 2020)
  - Data Analysis (2016, 2017)

Computer and natural languages

- Python, PyTorch/Tensorflow, C++, Java, Mathematica, Fortran
- Dutch (Native), English (Full professional proficiency)